

Wednesday Warm Up: Unit 4, Transformers and Inductors

Prof. Jordan C. Hanson

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1 Memory Bank

- $\epsilon = -N\Delta\phi_m/\Delta t$... Faraday's Law, with flux ϕ
- $\epsilon = -L\Delta I/\Delta t$... Faraday's Law, with inductance L
- $\epsilon_2/\epsilon_1 = N_2/N_1$... Transformer equation
- $\epsilon(t) = \epsilon_0(2\pi f)\cos(2\pi ft)$... Variation of induced voltage as a function of time and frequency.
- $U = \frac{1}{2}LI^2$... Energy stored in an inductor.

2 Faraday's Law, Transformers, and Scaling Properties

1. Consider Fig. 1, which depicts a *transformer*. There are two solenoids, the *transmit* and *receive* coils. Using Faraday's law at constant frequency, convince yourself that

$$\frac{\epsilon_{TX}}{\epsilon_{RX}} = \frac{N_{TX}}{N_{RX}} \quad (1)$$

2. A battery charger meant for a series connection of ten nickel-cadmium batteries (total emf of 12.5 V DC) needs to have a 15.0 V output to charge the batteries. It uses a step-down transformer with a 200-loop primary and a 120 V input. (a) How many loops should there be in the secondary coil? (b) If the input current is 16.0 A, what is the output current?

3. Suppose Fig. 1 represents our Faraday's Law lab activity (using an iron core to connect the flux rather than overlapping the coils). (a) If we observe $\epsilon_{RX}(t) = 5.0\cos(2\pi ft)$ Volts, with $f = 1$ kHz, what will $\epsilon_{RX}(t)$ be at $f = 5$ kHz? (b) What is the period of $\epsilon_{RX}(t)$ when $f = 5$ kHz?

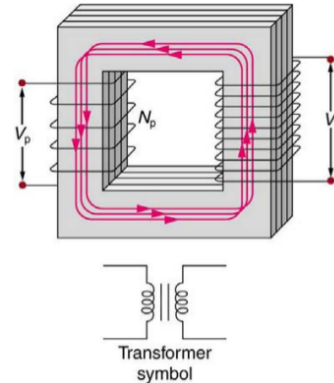


Figure 1: A basic diagram of a transformer with transmit and receive coils.

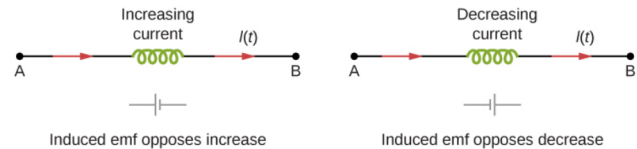


Figure 2: Inductors and induced voltage.

3 Inductors

1. Show that 1 H is equal to 1 $\Omega \cdot s$.
2. Camera flashes charge a capacitor to high voltage by switching the current through an inductor on and off rapidly. In what time must the 0.100 A current through a 2.00 mH inductor be switched on or off to induce a 500 V emf?
3. What is the energy stored in the inductor in the previous exercise, at 0.1 A?